Resilience Academy

SDI Training for ICT Professionals

Technology Overview

The tools behind a Spatial Data Infrastructure

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Day 1 | 1 Hour | crd.resilienceacademy.ac.tz

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Resilience Academy

Session Overview

What we will cover:

1. GeoNode — the platform that powers CRD
2. How the pieces fit together (simplified architecture)
3. Docker — why we use it and what it means for you
4. The three GUI tools you will use daily
5. Hardware requirements

Goal: understand the *big picture* so the hands-on sessions make sense. You will **not** be asked to configure any of this yourself.

Key Message

You do not need to become a systems administrator. Think of this session as looking under the hood of a car — it helps you understand what is happening, even though you will mostly just drive.

GeoNode — "WordPress for Maps"

What is GeoNode?

GeoNode is a free, open-source web platform for managing and sharing spatial data. Think of it as **WordPress, but instead of blog posts you publish maps and datasets**.

What GeoNode lets you do (via a browser):

- **Upload** spatial data (Shapefiles, GeoTIFFs, etc.)
- **Search** and browse a data catalogue
- **Preview** maps in the browser — no software needed
- **Download** data in multiple formats
- **Set permissions** — control who can view or

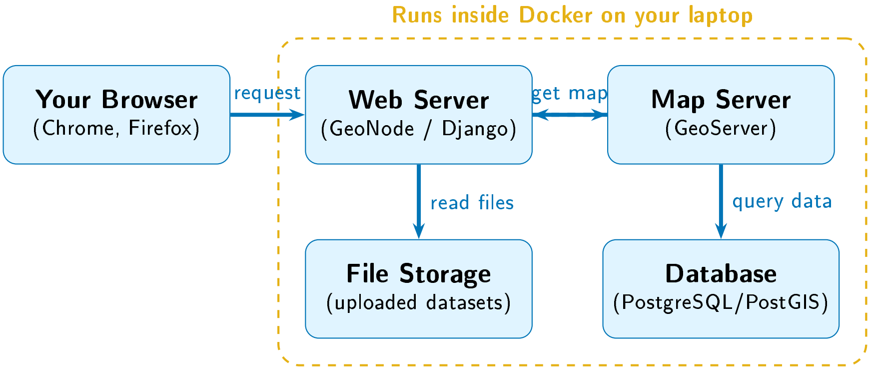
The WordPress Analogy

WordPress	GeoNode
Blog post	Dataset layer
Page	Web map
Media library	Data catalogue
Theme/plugin	Map style
Admin panel	GeoNode admin

CRD runs on GeoNode

Everything you saw on **crd.resilienceacademy.ac.tz** is powered by GeoNode. During this training you will

How the Pieces Fit Together



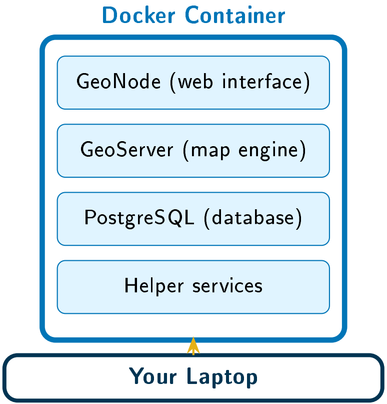
You only interact with the browser. Everything inside the dashed box runs automatically. Docker keeps all those pieces packaged together.

The **map server** speaks OGC standards (WMS, WFS, WCS) — the same ones CRD uses. Your local GeoNode works exactly like the real thing.

Docker — The "Shipping Container" for Software

The Problem:
GeoNode needs a web server, a map server, a database, and several helper programs. Installing all of them by hand is complicated and error-prone.

The Solution — Docker:
Docker packages *everything* GeoNode needs into a virtual "shipping container" that runs the same on any computer.



The Shipping Container Analogy

A real shipping container carries goods from China to Dar es Salaam. The ship does not care what is inside — it just moves the container

What This Means for You

You install **Docker Desktop** (a GUI app)

Three Tools You Will Use Every Day

QGIS

Desktop GIS

What it is:

A free application for viewing, editing, and analysing spatial data on your computer.

Think of it as:

"Google Earth Pro on steroids" — but free and far more powerful.

You will use it to:

- View downloaded datasets
- Style and label maps

Web Browser

GeoNode / CRD

What it is:

Chrome or Firefox — the tool you already know.

You will use it to:

- Browse and search the CRD catalogue
- Upload datasets to GeoNode
- Edit metadata
- Preview and share web maps

KoboCollect

Mobile Data Collection

What it is:

A free Android app for collecting field data with GPS coordinates.

Think of it as:

A digital survey form that automatically records your location.

You will use it to:

- Collect point data in the field
- Attach photos and GPS

Hardware Requirements

For this training (laptop):

- **RAM:** 8 GB minimum, 16 GB recommended
  Docker + GeoNode use about 4–6 GB
- **Disk space:** At least 20 GB free
  Docker images are large (~5 GB), plus your datasets
- **Processor:** Any modern Intel/AMD or Apple M-series
  Must support virtualisation (most laptops from 2018+ do)
- **OS:** Windows 10/11 (with WSL2), macOS, or Ubuntu Linux

For a Real SDI Server

A production GeoNode (like CRD) runs on a cloud server with:

- 16–32 GB RAM
- 4–8 CPU cores
- 500 GB+ storage
- Reliable internet connection
- A domain name (e.g. crd.resilienceacademy.ac.tz)

You do *not* need this for the training — your laptop is enough.

Session 2 Summary

Key takeaways:

- **GeoNode** is the "WordPress for maps" — an open-source platform that powers CRD
- The architecture has four layers: browser, web server, map server, database — but **you only use the browser**
- **Docker** packages everything into a portable container so setup is simple and consistent
- Your three daily tools are **QGIS** (desktop maps), a **browser** (GeoNode/CRD), and **KoboCollect** (field data)

Remember

- You do **not** need to be a programmer
- You do **not** need to configure servers
- You **do** need to understand what each tool does, so you can troubleshoot basic issues and talk to your IT team

Next Session

Session 3: Linux Basics

We learn 10 essential "recipes" for the terminal — just enough to navigate files and run commands when needed.